

THE UNIVERSITY OF HONG KONG
HKU Business School
Master of Science in Business Analytics
MSBA7030 Digital Innovation

I. General Information

Class time: Please refer to Teaching Plan
Class venue: Please refer to Teaching Plan
Instructor: Prof. Jack Jiang
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Office: KKL 804
Phone: 3917 8351
Consultation times: By appointment

Tutor: TBC

II. Course Description

Information technology (IT) enables new patterns of commerce and affects how buyers buy and sellers sell, how firms interface with their customers and suppliers, and how activities are organized internally within the firm, and how individuals utilize technologies. The integration of technology, individuals, and business processes makes it pertinent for the managers and leaders of today and tomorrow to be able to understand and communicate the strategic potential of design and leverage digital innovation for business competition. We will focus on innovations in technology-driven business models. The course contents will cover contemporary topics on innovation strategies, platform economics, business model development, disruptive technologies, digital experiments, cryptocurrency, artificial intelligence, information privacy, etc. In particular, students will learn how to identify innovation opportunities and manage innovation processes. They will also be able to master the ability to develop digital business models.

III. Course Objectives

This course has the following objectives:

- Provide students with a solid theoretical understanding of digital innovation and digital business models;
- Foster an understanding of the essential principles of IT management as a critical component of organizational competitiveness;
- Explore new innovations enabled by digital technologies;
- Instruct students on evaluating the role of emerging technologies in business and market practices.
- Provide fundamental training on digital experiments and A/B testing.

IV. Course Content and Tentative Teaching Schedule

This will include a schedule of tentative topics, readings, continuous assessment and other activities for each week during the course.

Date	Instructional Period:	Lecture Topics	Special Notes
	Session 1	♦ Business Models	Case: WeChat (assignment due)
	Session 2	♦ Platform Competition	Case: Amazon (assignment due)
	Session 3	♦ Long Tail and Product Recommender	Case: ByteDance (assignment due)
	Session 4	♦ Experimental Evaluation	
	Session 5	♦ Cryptocurrency	Case: Visa (assignment due)
	Session 6	♦ Metaverse: Opportunities and Challenges	In-class Quiz Case: Meta (assignment due)
	Session 7	♦ Generative AI (I)	
	Session 8	♦ Generative AI (II)	Guest talk
	Session 9	♦ Generative AI (III) ♦ Privacy and Security	
	Session 10	♦ Privacy and Privacy Computing	Case: Apple (assignment due) In-class Quiz
			Project due and presentation

V. Specific Arrangements

Course Evaluation (100% in total):

1. Class participation and presentation (30%)
2. Individual Assignment and Quizzes (40%)
3. Project (30%, including group work 20% and individual contribution 10%)

Course Design:

The design of this course comprises *lectures* and *case discussion*, both of which are highly interactive. Students are required to spend substantial time studying the prescribed readings before attending class. Students are also expected to participate and be engaged in all class activities (including normal lectures and class presentations).

Note: *Each student should bring a name tent to the class to facilitate communication and discussion.*

Cases:

Throughout the course, you will be exposed to a number of theoretical frameworks and disciplinary perspectives. Cases allow us to ground these theories in the context of real business scenarios. Cases are carefully selected to cover a range of industries and business models over the past decade. The objective is for you to generalize the theories and principles across contexts in which information technologies and digitization are changing the nature of the business. Considerable emphasis is placed on emerging digital business practices. You may be required to present relevant cases.

Before each session, you are not only required to get familiar with the basic concepts in the readings, but more importantly, you need to read cases and THINK critically and creatively along key questions of the cases. You are also expected to actively discuss, debate, and challenge each other during the class.

Six cases will be discussed in the class. You are required to submit two case assignments, addressing specific discussion questions. These questions can be accessed on Moodle. The allocation of case assignment will be done randomly. Please submit your answer to the case questions prior to the corresponding class session (except for the first case). Please limit your answer to 1.5 pages.

Quizzes:

In-class quizzes will be conducted in Session 6 and Session 10 to assess students' understanding of the course contents.

Group Project:

You can pick one of the following two options:

- 1) Propose a business plan that involves digital innovation, which can be either IT-based product innovation, service innovation, platform innovation, or a combination of these innovations. You are encouraged to generate creative and novel ideas about products or services and to tap an untapped market.
- 2) Write a case based on a real firm's digital innovation. You must have very thorough understanding of internal operations and strategies of the firm. Special approval is needed from the lecturer if you want to take Option 2.

For either option, students will work in groups to complete the term paper. You are expected to report an in-depth analysis of the industry in which the firm is operated, and to develop (for option 1) or to elaborate (for option 2) a digital innovation plan that allows the firm to gain competitive advantage and sustainable success. Specifically, you need to:

- study the industry and the critical success factors for that industry. Also, you will study the firm, its business, customers, and suppliers. You need to propose and analyze the firm's innovation strategies, and the specific business processes the firm should focus on in order to stay competitive;
- write a business plan by focusing on how IT can facilitate innovation and create competitive advantages.

Qualitative, as well as quantitative, analyses are required. The information sources for the project will determine to a great extent the quality of the term paper; therefore, it is highly recommended that you write your report based on extensive surveys, such as various industrial reports.

We are very strict about double submission. Submitting the same project idea or similar project ideas to two or more different modules is strictly prohibited. The lecturers will frequently communicate with each other to ensure there is absolutely no double submission. If any case is identified, disciplinary action will be taken immediately.

VI. Course Policy:

1. ***Late assignment/project submission will NOT be accepted.*** The exceptions are doctor's note, prior approval by the instructor for special circumstances (e.g. representing HKU on a one-week sports competition).
2. For individual assignment, each student should complete the work on their own. If two projects are judged by the marker to be highly similar or identical, a grade of zero will be given to both students involved.
3. Students are encouraged to search relevant materials from external sources, such as journals, newspapers, and the Internet. Wherever facts, ideas, or figures from other sources are used, you need to clearly indicate the original sources and put them in a reference list. Plagiarism is strictly prohibited. Each student will be responsible for the academic honesty of the entire group work. In other words, in case a plagiarism is found, the grades of all students belonging to the group will be zero.

Reading List
(subject to change)

Session 1: IT-Strategy and Business Models

1. Chapter 2 “Business Model Frameworks,” in *Business Model Innovation: Concepts, Analysis, and Cases*, by Afuah, A. Routledge, 2nd Edition, 2018. [\[available online\]](#)
2. Case: Huang, S., Yan, S., Jiang, Z., and Huang, M. “ESG at WeChat Pay to Support SMES,” Asia Case Research Centre, HKU Business School, 2022 [\[to be purchased\]](#)

Optional:

3. Christensen, C. M., Raynor, M. E. and McDonald, R. “What is Disruptive Innovation?” *Harvard Business Review*, Vol. 93, No. 12, 2015, pp. 44-53. [\[available online\]](#)

Session 2: Platform Competition

4. Chapter 2 “Network Effects: The Power of the Platform,” in *Platform Revolution: How Networked Markets Are Transforming the Economy and How to Make Them Work for You*, by Parker, G. G., Van Alstyne, M. W. and Choudary S. P., W. W. Norton & Company, 2016, pp. 21-35. [\[available online\]](#)
5. Van Alstyne, M. W., Parker, G. G., and Choudary, S. P. “Pipelines, Platforms, and the New Rules of Strategy,” *Harvard Business Review*, Vol. 94, No. 4, 2016, pp. 54-62. [\[available online\]](#)
6. Case: Rothaermel, F. T. “Amazon.com, Inc.,” McGraw Hill Education, MH0068, 2019. [\[to be purchased\]](#)

Session 3: Long Tail and Product Recommender

7. Chapter 2 “Laying the Tracks: Digital and Socioeconomic Foundations,” in *The Sharing Economy: The End of Employment and the Rise of Crowd-Based Capital*, by Sundararajan, A. The MIT Press, 2016, pp. 46-67. [\[available online\]](#)
8. Brynjolfsson, B., Hu, Y. J. and Smith, M., “From Niches to Riches: Anatomy of the Long Tail,” *MIT Sloan Management Review*, Vol. 47, No. 4, Summer 2006. [\[available online\]](#)
9. Boudreau, K. J. and Lakhani, K. R. “Using the Crowd as an Innovation Partner,” *Harvard Business Review*, Vol. 91, No. 4, 2013. [\[available online\]](#)
10. Case: Davis, J. P., Yang, A., Vo, M., “TikTok's AI Strategy: ByteDance's Global Ambitions,” INSEAD, Publication Date: May 2019 (Revised Oct 2020), Product #: IN1587-PDF-ENG. [\[to be purchased\]](#)

Session 4: Experimental Evaluation

11. Chapter 10 “Experimental Research” in *Social Science Research: Principles, Methods, and Practices*, by Anol Bhattacharjee, 2nd edition, 2012. [\[available online\]](#)
12. Lee, D. and Hosanagar, K. “How Do Recommender Systems Affect Sales Diversity? A Cross-Category Investigation via Randomized Field Experiment,” *Information Systems Research*, Vol. 30, No. 1, 2019, pp. 239-259. [\[available online using HKU library authentication\]](#)
13. Liu, Y., Jiang, Z. and Choi, B. “Pushing Yourself Harder: The Effects of Touch Modes on Users’ Self-Regulation,” *Information Systems Research*, Vol. 34, No. 3, 2023, pp. 996-1016. [\[available online\]](#)

Session 5: Cryptocurrency

14. Nakamoto, S. "Bitcoin: A Peer-to-Peer Electronic Cash System," <https://bitcoin.org/bitcoin.pdf>
15. Ramsey Khabbaz, "Web3 Will Run on Cryptocurrency," Harvard Business Review, Product #: H070U0-PDF-ENG, May 2022, [\[available online\]](#)
16. Case: Agnihotri, A. and Bhattacharya, S. "Visa Inc.: Threat from Cryptocurrency?" IVEY Publishing, 2020. [\[to be purchased\]](#)

Session 6: Metaverse: Opportunities and Challenges

17. Park, S. M., & Kim, Y. G. (2022). A Metaverse: Taxonomy, components, applications, and open challenges. *Ieee Access*, 10, 4209-4251.
<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9667507>
18. Balis, J. "How Brands Can Enter the Metaverse," Harvard Business Review, 2022. [\[available online using HKU Library authentication\]](#)
19. Kaczynski, S. and Kominers, S. D. "How NFTs Create Value," Harvard Business Review, 2021. [\[available online using HKU Library authentication\]](#)
20. Case: Kim, W. C., Mauborgne, R. and Olenick, M. "Meta: Facebook's Pivot to the Metaverse. A Path to Dystopia or Blue Ocean Utopia?" INSEAD Publishing, 2022. [\[to be purchased\]](#)

Session 7: Generative AI (I)

21. Wolfram, Stephen, *What Is ChatGPT Doing ... and Why Does It Work?*, Wolfram Research, Inc., 2023. (<https://writings.stephenwolfram.com/2023/02/what-is-chatgpt-doing-and-why-does-it-work/>)
22. Appel, G., Neelbauer, J. and Schweidel, D.A. *Generative AI Has an Intellectual Property Problem*, Harvard Business Review, April, 07, 2023. [\[available online\]](#)

Session 8: Generative AI (II)

23. David C. Edelman and Mark Abraham, "Generative AI Will Change Your Business. Here's How to Adapt," Harvard Business Review, 2023 April, [\[available online using HKU Library authentication\]](#)
24. Tojin T. Eapen, Daniel J. Finkenshtadt, Josh Folk, and Lokesh Venkataswamy, "How Generative AI Can Augment Human Creativity," Harvard Business Review, 2023 July - August, [\[available online using HKU Library authentication\]](#)
25. Oguz A. Acar, "AI Prompt Engineering Isn't the Future," Harvard Business Review, 2023 June, [\[available online using HKU Library authentication\]](#)
26. Kathy Baxter and Yoav Schlesinger, "Managing the Risks of Generative AI," Harvard Business Review, 2023 June [\[available online using HKU Library authentication\]](#)

Session 9: Privacy and Security

27. Acquisti, A., Brandimarte, L. and Loewenstein, G., 2015. Privacy and human behavior in the age of information. *Science*, 347(6221), pp.509-514. [\[available online\]](#)
28. Case: McGee, Henry W., Nien-hê Hsieh, Sarah McAra, and Christian Godwin. "Apple: Privacy vs. Safety (A)." Harvard Business School Case, Product #: 321004-PDF-ENG, February 2021. [\[to be purchased\]](#)

Session 10: Privacy and Privacy Computing

29. "The Next Generation of Data-Sharing in Financial Services: Using Privacy Enhancing Techniques to Unlock New Value," White Paper, World Economic Forum, 2019, [\[available online\]](#)